

IILM University

Greater Noida, Uttar Pradesh

SQL and Database Management Systems

AICTE – EduSkills Virtual Internship | April – June 2026

Student Name	Varun Tomar
Enrollment Number	CS-2341311
Program & Semester	B.Tech CSE AIML & 7th Semester
Organization	EduSkills Academy (AICTE)

Internship Overview

8 Weeks	Virtual	Intern	SQL & DBMS
Apr – Jun 2026	AICTE – EduSkills	Role	Domain

Organization Profile

EduSkills Academy
Domain: EdTech & Skill Development
Supported by AICTE; under the patronage of the National Internship Portal
ISO certified | Great Place To Work certified

Evaluation Structure

- Week-wise structured learning plan
- Weekly assessments
- Project documentation & assignments
- Final Assessment Test in the last week
- Grade from overall performance

Problem Statement & Objectives

Problem Statement

Organizations generate large volumes of data. It must be stored in well-designed databases, queried efficiently, secured, integrated with applications through APIs, and presented as insights. This internship connects all of these steps, ending with a data-driven business analytics dashboard.

Objectives

1. Design relational databases and write advanced SQL
2. Work with NoSQL (MongoDB) and ETL
3. Visualize data with Power BI
4. Build APIs with Node.js/Express & FastAPI
5. Apply performance, security & backup practices

8-Week Learning Roadmap

1 Week 1

Database Foundations & Relational Design

ER modeling, normalization, DDL/DML

2 Week 2

Advanced SQL & MongoDB

Joins, subqueries, GROUP BY, CRUD

3 Week 3

Aggregation, ETL & Power BI

ETL, transformations, first dashboards

4 Week 4

Data Modeling & Node.js Integration

DAX, PostgreSQL, Express CRUD APIs

5 Week 5

Backend APIs with FastAPI

SQLAlchemy, REST APIs, requests

6 Week 6

Performance, Security & Backup

Indexing, roles, SQL injection, backup

7 Week 7

Scalability & High Availability

Replication, sharding, case studies

8 Week 8

Advanced NoSQL & Capstone

Aggregation pipeline, time series, dashboard

Technologies & System Architecture



Data Sources

Datasets, files, REST APIs

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ETL

Extract, clean, transform, load

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Databases

PostgreSQL (SQL) | MongoDB (NoSQL)

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API Layer

Node.js/Express, FastAPI/SQLAlchemy

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Power BI

Data model, DAX, dashboard

Tools & Concepts

SQL | PostgreSQL | MongoDB | Power BI & DAX | Node.js & Express | Python & FastAPI

SQLAlchemy | REST APIs | Indexing | Replication & Sharding

Capstone: Business Analytics Dashboard

1. **Collect & Clean** Extract data with SQL and ETL, fix quality issues
2. **Store & Combine** Load into databases; merge API and other datasets
3. **Model** Power BI relationships and DAX measures
4. **Visualize** Interactive dashboard for decision-making
5. **Extend with NoSQL** MongoDB aggregation, indexing, time series

Security & Performance

- Indexing & query optimization
- Roles and permissions
- SQL injection prevention
- Backup & recovery
- Replication, sharding, clustering

Learning Outcomes

1 Database Design

Normalized relational schemas with ER modeling and SQL

2 Querying

Advanced SQL and MongoDB aggregation pipelines

3 Data Analytics

ETL, Power BI data models, DAX and dashboards

4 Integration

RESTful CRUD APIs with Node.js/Express and FastAPI

5 Security & Scale

Performance tuning, access control, backup, high availability

6 End-to-End Delivery

A data-driven analytics dashboard as capstone

Conclusion & Future Scope

The internship gave a complete view of database management: from designing a relational schema to delivering business insights through a dashboard, using both SQL and NoSQL systems.

Future Scope

- Cloud-hosted and real-time data pipelines
- Automated ETL with scheduled dashboard refresh
- Production-style replication, sharding and backup

Thank You